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Design of Two-Stage Operational Amplifier using Indirect Feedback Frequency Compensation

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Design of Two-Stage Operational Amplifier using Indirect Feedback Frequency Compensation

Design of Two-Stage Operational Amplifier using Indirect Feedback Frequency Compensation

An undergraduate Honors thesis submitted in partial fulfilment
of the requirements for the degree of
Bachelor of Science in Electrical Engineering

by
Roderick A. Gomez

May 2019
University of Arkansas

Abstract

This thesis work details the designing process of two silicon two-stage operational amplifiers with indirect feedback compensation and with Miller compensation technique. The main objective of this thesis is to study the advantages of indirect feedback compensation in comparison with Miller compensation and how this technique can be applied to meet certain design specifications. The operational amplifiers are designed with 130 nm Silicon Germanium CMOS process ideally for temperature range of 25°C to 300°C. The two op-amps are designed to have a DC gain of about 70 dB and 60 degrees of phase margin. The indirect feedback compensation design showed similar simulation results as the Miller compensation technique; nevertheless, it showed a reduce in the compensation capacitor size, meaning a smaller design area, and an improvement in the phase margin from the LHP zero. Also, the proposed design showed a higher unity gain frequency. Further analysis of indirect feedback frequency compensation on multistage amplifiers (greater than two) should be conducted to analyze the potential of this compensation method under more complex compensation against the commonly used Miller technique.

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Introduction

The purpose of this thesis is to report the design procedures of a two-stage operational amplifier with indirect feedback compensation. This compensation method is not widely used in operational amplifiers; however, its application on frequency compensation can help improve the design performance of op-amp. The report will cover the main differences between this method and the common direct compensation or Miller compensation, and the advantages and disadvantages of indirect feedback compensation.

This document is divided into 2 sections. First, the background where all the theory behind frequency compensation is explained. This will include the concepts behind each frequency compensation method and how indirect feedback compensation presents a benefit for the design of operational amplifiers. Secondly, the design process for a two-stage operational amplifier with miller capacitor compensation and the design process for a two-stage operational amplifier with indirect feedback compensation. The two designs will be based on the same design specifications to make a comparison. The design simulations and discussion will cover the performance of each amplifier and explain how the indirect feedback compensation results shows an improvement in certain aspect of the operational amplifier design.

Background and Conceptual Principles

CMOS operational amplifiers are one of the most fundamental, versatile and integral building blocks of many analog and mixed-signal circuits and system. They are used in a wide range of applications such as comparators, differentiators, dc bias applications and many other applications. IC designers tend to design systems with a single dominated pole behavior because these are easily analyzed and can tolerate negative feedback without stability issues. As a result,

single stage operational amplifiers have been preferred for their stable frequency response. However, CMOS technology has been constantly scaling down establishing some challenges when designing operational amplifiers and others integrated circuits. Additionally, the power supply voltage has also been reduced, causing techniques like cascading of transistors more difficult to implement. The new scaled processes enable faster speeds, but lower open loop gains and the reduction in voltage does not allow for cascading multiple stages to achieve higher gains. Therefore, alternative architectures must be implemented to overcome the drawback of single stage amplifiers. Multiple stage amplifiers can be implemented to achieve higher gains circuit designs regardless of the limitations of the power supply voltage and other performance aspects that affect single stage amplifiers. However, multiple stage amplifiers are generally complex to compensate. Two-stage operational amplifiers are the most common used multistage amplifier because it can provide high gain and high output swing. However, an uncompensated two-stage operational amplifier has a two-pole transfer function, and these are located below the unity gain frequency. Therefore, a frequency compensation circuitry must be implemented to ensure stability. It is difficult to design a system with a truly single pole behavior; nevertheless, this desire behavior can be approximate over a frequency range that falls under the desire design specifications.

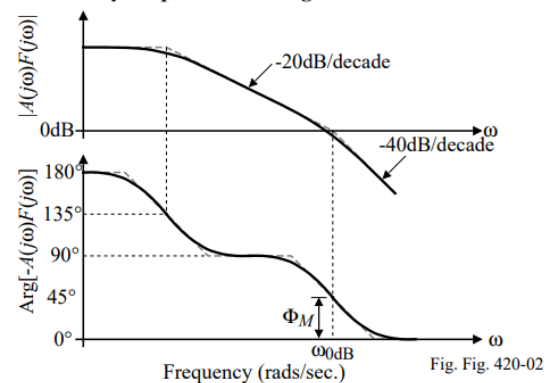
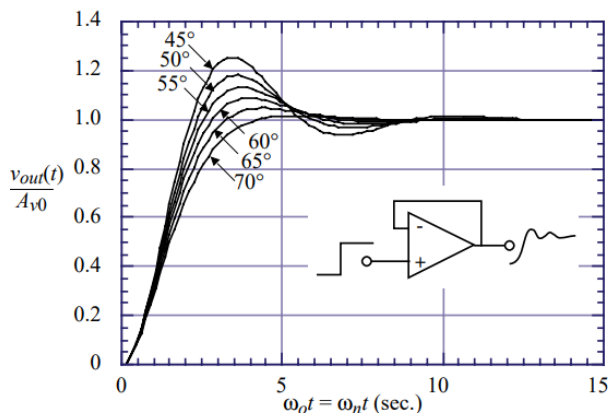


Figure 1. Stability problem on an amplifier and how it is important for the step response [1]

Operational amplifiers operated on a close-loop with a negative-feedback system are susceptible to oscillation. The measurement of stability of an operational amplifier is the phase angle at unity open-loop gain and this is given by [1]

$$Phase\ Margin = \Phi_M = Arg[-A(j\omega_{0,dB})F(j\omega_{0,dB})] = Arg[L(j\omega_{0,dB})] \quad (1.1)$$

where the negative feedback is illustrated as follow.

$$L(s) = -A(s)F(s): Open - Loop\ Gain \quad (1.2)$$

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{A(s)}{1 + A(s)F(s)}: Closed - Loop\ Gain \quad (1.3)$$

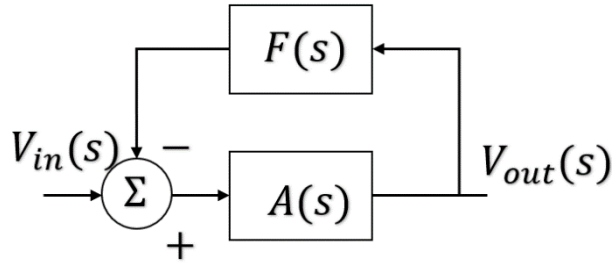


Figure 2. Block Diagram of feedback configuration

Due to the parasitic components on the amplifier, in addition to attenuation there is a phase shift between input and output, and oscillations will happen when the phase shift (phase margin) exceeds 180 degrees. A phase margin of 180 degrees turns negative feedback into positive feedback causing the amplifier to oscillate. As a result, the more stages an amplifier has, the more unstable its behavior is, requiring more complex compensation methods. As a rule of thumb, a 45 degree or greater is a phase margin that will yield good stability and less overshoot [1]. Furthermore, as shown on figure 1, stability is important in order to have a good step response on the amplifier. The desired behavior of an amplifier is to reach its final value quickly; therefore, the amplifier must be stable and have a phase margin at least greater than 45 degrees.

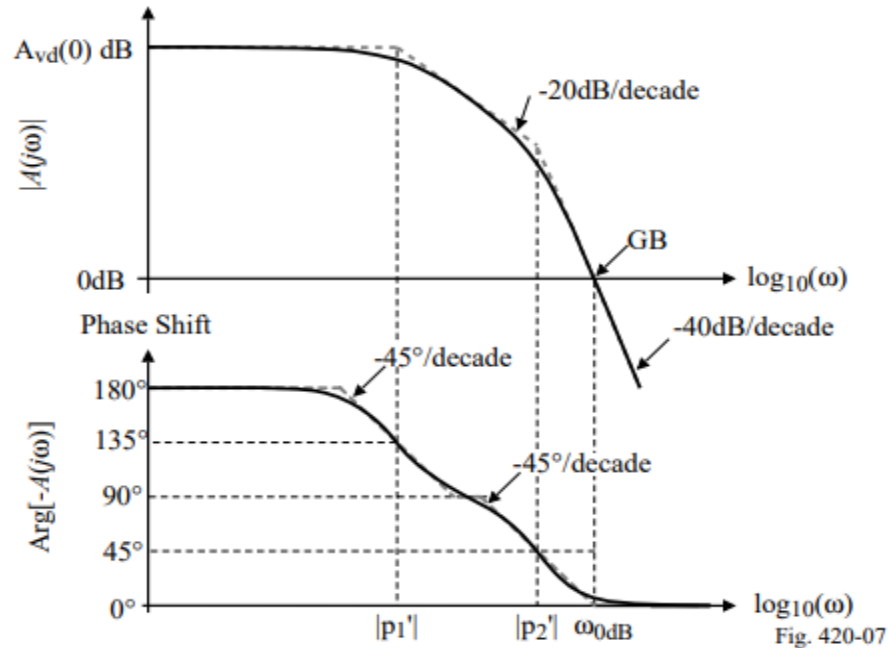


Figure 3. Frequency Response of an uncompensated operational amplifier [1]

A two-stage operational amplifier consists of a differential amplifier at the input stage, while the second stage is a high gain stage biased by the output of the differential amplifier. As explained before, two-stage operational amplifier exhibits two poles below the unity open-loop gain. As shown on figure 3, when the gain of the two-stage operational amplifier is equal to the unity gain frequency, the phase shift is less than 45 degrees. Therefore, to achieve stability, a two-stage operational amplifier must be compensated. The most widely used compensation architecture in analog circuit and system design is pole splitting using the Miller effect. This is known as the Miller compensation technique.

Miller Compensation Technique Principles

The Miller effect makes one pole more dominant by moving the pole down in frequency, while the other becomes less dominant by moving the pole up in frequency (pole splitting). This action is intended to achieve adequate phase margin by forcing the system transfer function to behave like a single pole system. The Miller compensation technique consists

on a compensation capacitor placed between the output of the first stage (differential amplifier) and the output of the operational amplifier (output of the gain stage amplifier). A block diagram is shown on figure 4.

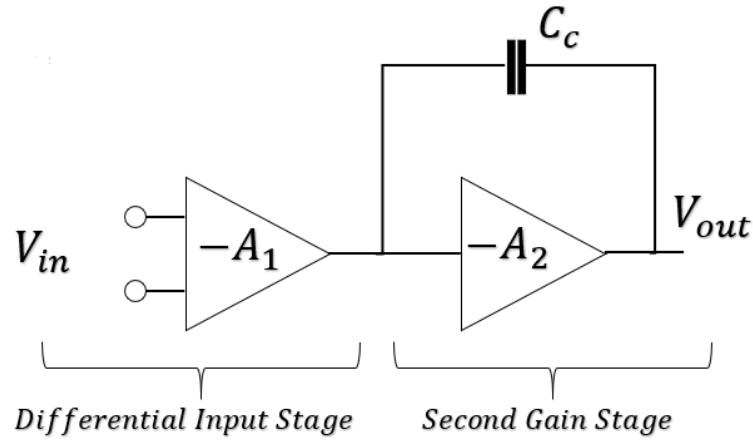


Figure 4. Block diagram of a Miller compensated operational amplifier

The transfer function for a Miller compensation two-stage operational amplifier with small signal model shown on figure is computed as follow

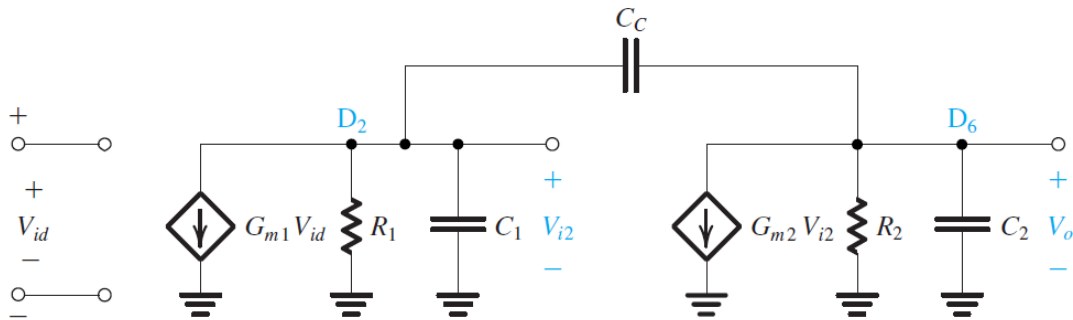


Figure 5. Small signal model of the Miller compensated operational amplifier [2]

$$\frac{V_{i2}}{\frac{1}{sC_1}} + \frac{V_1}{R_1} + G_{m1}V_{id} + \frac{V_{i2} - V_o}{\frac{1}{sC_c}} = 0 \quad (2.1)$$

$$\frac{V_o}{\frac{1}{sC_2}} + \frac{V_o}{R_2} + G_{m2}V_{i2} + \frac{V_o - V_{i2}}{\frac{1}{sC_c}} = 0 \quad (2.2)$$

$$\frac{V_o(s)}{V_{id}(s)} = \frac{G_{m1}R_1G_{m2}R_2 \left(1 - s\frac{C_c}{G_{m2}}\right)}{s^2[R_1R_2(C_1C_2 + C_1C_c + C_2C_c)] + s[R_1(C_1 + C_c) + R_2(C_2 + C_c) + G_{m2}R_1R_2C_c] + 1} \quad (2.3)$$

$$\frac{V_o(s)}{V_{id}(s)} = \frac{A_{DC} \left(1 - \frac{s}{z_1}\right)}{\left(1 - \frac{s}{p_1}\right)\left(1 - \frac{s}{p_2}\right)} \quad (2.4)$$

$$z_1 = \frac{G_{m2}}{C_c} \quad (2.5)$$

$$p_1 \cong -\frac{1}{G_{m2}R_1R_2C_c} \quad (2.6)$$

$$p_2 \cong -\frac{G_{m2}C_c}{C_1C_2 + C_1C_c + C_2C_c} \cong -\frac{G_{m2}}{C_1 + C_2} \quad (2.7)$$

Without a compensation capacitor between the output of the first stage and the output of the second stage, the two poles of the two-stage operational amplifier are given as [3]

$$p_1 = \frac{1}{R_1C_1} \quad (3.1)$$

$$p_2 = \frac{1}{R_2C_2} \quad (3.2)$$

As a consequence of the Miller compensation technique, a right half-plane (RPH) zero is introduced in the two-stage operational amplifier due to the feed-forward current from the output of the first stage and the operational amplifier output since the Miller effect can increase significantly the time constant related to the compensation capacitor [4]. This is an undesirable effect because it degrades the phase margin limiting the maximum bandwidth of the two-stage operational amplifier. Due to these reasons, the compensation capacitor size is large on the two-stage op-amp.

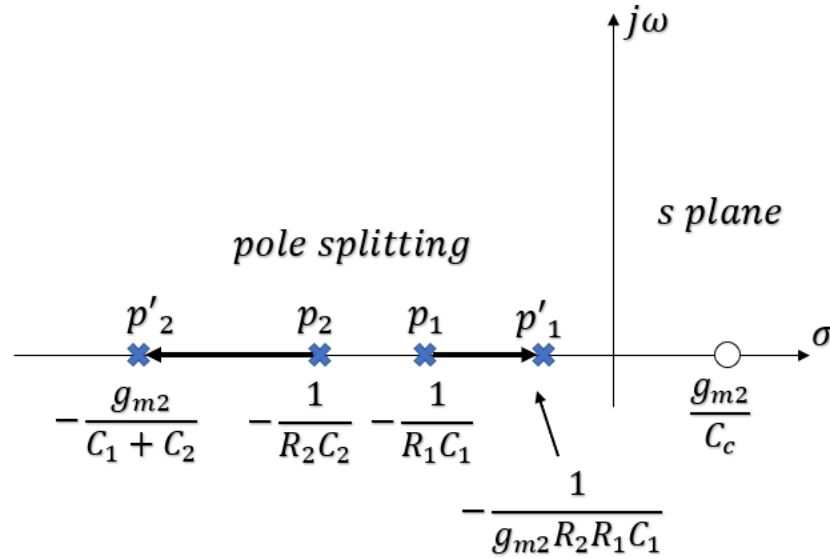


Figure 6. Pole-Zero plot of the Miller effect on the operational amplifier

As shown on the pole-zero plot, the poles of the input and output are split apart, thus achieving the dominant and non-dominant poles, which result in the system behaving as a first-order system. Many advanced techniques have been developed to overcome the drawback of the RHP zero introduced by the Miller effect. For example, nulling resistor miller compensation [5], active miller compensation [6] and voltage buffer type miller compensation [7] are examples of advanced frequency compensation techniques. As introduced in the last section, this thesis will explore the advantages of using indirect feedback compensation to split the two-pole system of the two-stage operational amplifier thus obtaining a single pole system.

Indirect Feedback Compensation Technique Principles

Indirect feedback frequency compensation is achieved by feeding the feedback current indirectly from the output to the internal high impedance node of the first stage [4]. In this frequency compensation method, the compensation capacitor is placed at a low impedance node in the first stage (differential amplifier) allowing indirect feedback current compensation from the output of

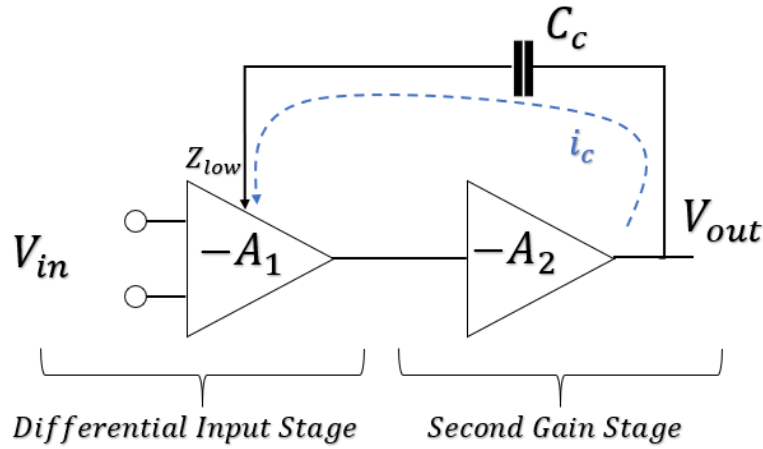


Figure 7. Block diagram of an indirect feedback compensated operational amplifier

the operational amplifier to the internal high impedance node of the output of the differential amplifier thus obtaining pole splitting and hence frequency compensation. Also, the right-hand plane (RPH) zero is eliminated by avoiding the direct connection of the compensation capacitor to the output of the differential amplifier. Besides the advantage of eliminating the RPH zero, the operational amplifier with indirect feedback compensation exhibits a significantly reduction in the layout [8].

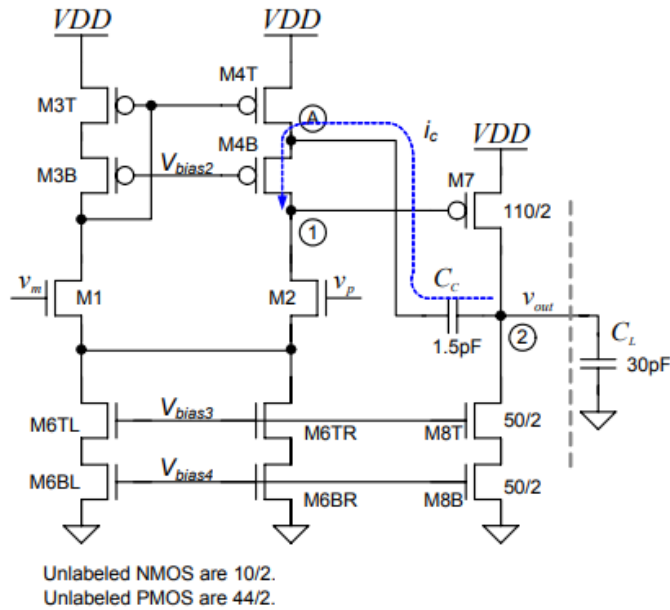


Figure 8. Schematic of two-stage operational amplifier with indirect feedback compensation [3]

The feedback current can be fed indirectly to the high impedance node of the differential amplifier using a cascode structure, using a common gate amplifier [9] or using a low impedance node of MOSFET laid out in series where one operates in a triode region. Figure 8 shows a two-stage operational amplifier with indirect feedback compensation. A general analysis of the small signal of indirect feedback two-stage operational amplifiers is given as follows [3]

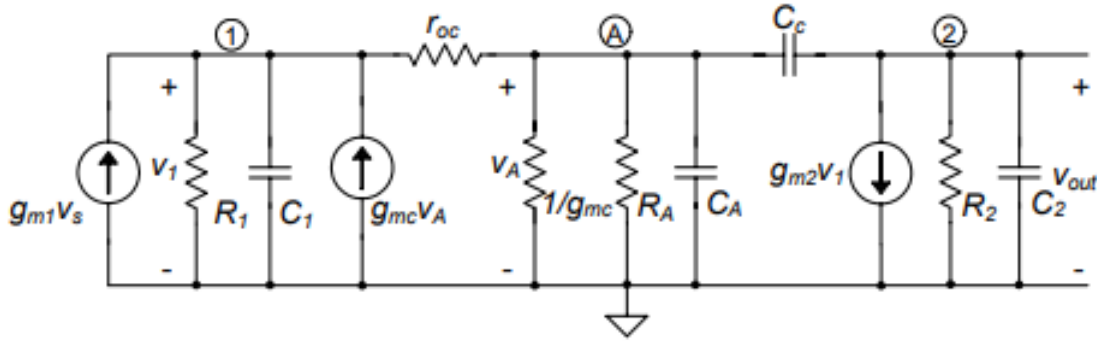


Figure 9. Small signal model of the indirect feedback compensated operational amplifier [3]

$$-g_{m1}V_s + \frac{V_1}{R_1} + V_1sC_1 - g_{mc}V_A + \frac{V_1 - V_A}{r_{oc}} = 0 \quad (4.1)$$

$$g_{m2}V_1 + \frac{V_{out}}{R_2} + V_{out}sC_2 + sC_c(V_{out} - V_A) = 0 \quad (4.2)$$

$$\frac{V_A - V_1}{r_{oc}} + g_{mc}V_A + V_AsC_A + \frac{V_A}{R_A} + sC_c(V_{out} - V_A) = 0 \quad (4.3)$$

$$\frac{v_{out}}{v_s} = -A_v \left(\frac{b_0 + b_1s}{a_0 + a_1s + a_2s^2 + a_3s^3} \right) \quad (4.4)$$

The transfer function of the two-stage amplifier with indirect frequency compensation consists of a real left-hand plane (LHP) zero and three poles.

The zero location is at

$$Z_1 \approx -\frac{g_{mc}}{C_c + C_A} \quad (4.5)$$

The three poles are located at

$$p_1 \approx -\frac{1}{g_{m2}R_2R_1C_C} \quad (4.6)$$

$$p_2 \approx -\frac{g_{m2}C_C}{C_1C_L} \quad (4.7)$$

$$p_3 \approx -\left[\frac{g_{mc}}{C_2||C_C} + \frac{1}{(R_1||r_{oc})C_1}\right] \quad (4.8)$$

By comparing the two equations for the non-dominant pole of the two-stage amplifier with Miller compensation $\left(-\frac{g_{m2}}{C_1+C_L}\right)$ and indirect feedback compensation $\left(-\frac{g_{m2}C_C}{C_1C_L}\right)$, it is clear that the second pole has moved further away from the first pole or the dominant pole by a factor of $\left(\frac{C_C}{C_1}\right)$. This fact implies that pole splitting can be achieved with lower value of compensation capacitor, meaning a higher unity gain frequency can be obtained without affecting the stability performance of the operational amplifier. From the transfer function, a LHP is introduced to the system which improve the phase margin. Also, as the compensation capacitor is smaller, the slew rate is improved. From the conceptual application of indirect feedback compensation, operational amplifiers with indirect frequency compensation can be designed with higher speed, lower power, and small layout area.

Two-Stage Operational Amplifier Design and Simulation

Design Specifications:

The following specifications will be used to design a two-stage operational amplifier with Miller compensation technique and indirect feedback frequency compensation to study the benefits of using indirect compensation as an alternative to the commonly used direct compensation.

Table 1. Required Design Specifications

Parameter	Value
DC Gain	70 dB
GBW	20 MHz
Phase Margin	$\geq 60^\circ$
Slew Rate	20 V/ μ s
V_{DD}	1.8 V
C_{Load}	2 pF
ICMR	0.6 V – 1.6 V
Power	$\leq 300 \mu W$

Design Process using Miller Compensation Technique:

The following design consist of an NMOS differential amplifier with active load as the first stage follow by a PMOS common source amplifier as the second stage. A compensation capacitor is connected between the output of the second stage and the output of the first stage to obtained pole splitting and hence op-amp compensation. Figure 10 shows the schematic implemented for the direct feedback two-stage op-amp design.

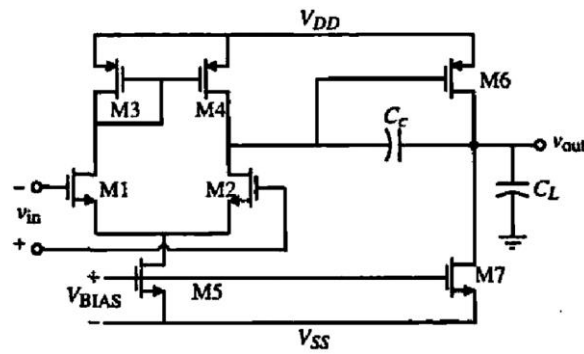


Figure 10. Schematic of a two-stage operational amplifier with Miller compensation.

Table II states the size of each transistor used in the Miller compensation two-stage operational amplifier.

Table 2. Transistor Sizing

Transistor	Aspect Ratio ($\frac{W}{L}$)
M1, M2	6
M3, M4	14
M5	12
M6	173
M7	75

Cadence Design and Simulation of Miller Compensation Amplifier

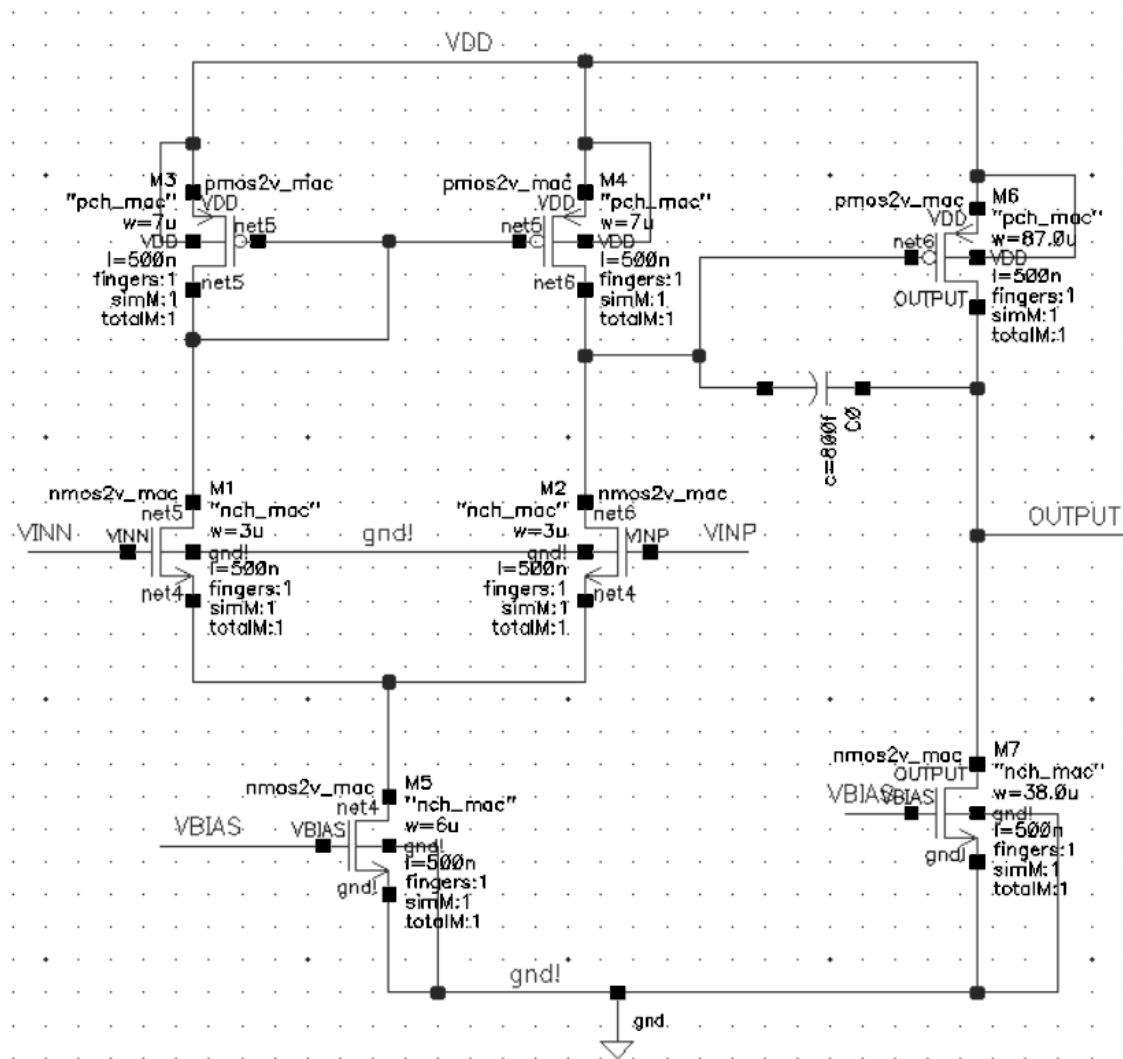


Figure 11. Design schematic of Miller compensated amplifier under analysis

The two-stage operation amplifier with Miller compensation achieved the desired specifications based on the sizing showed on Table II. As shown below on the frequency response of the operational amplifier, the system behaved as a single pole system before the unity gain frequency. This allows a better phase margin for the operational amplifier thus achieving better stability. Other simulations results are stated on Table III.

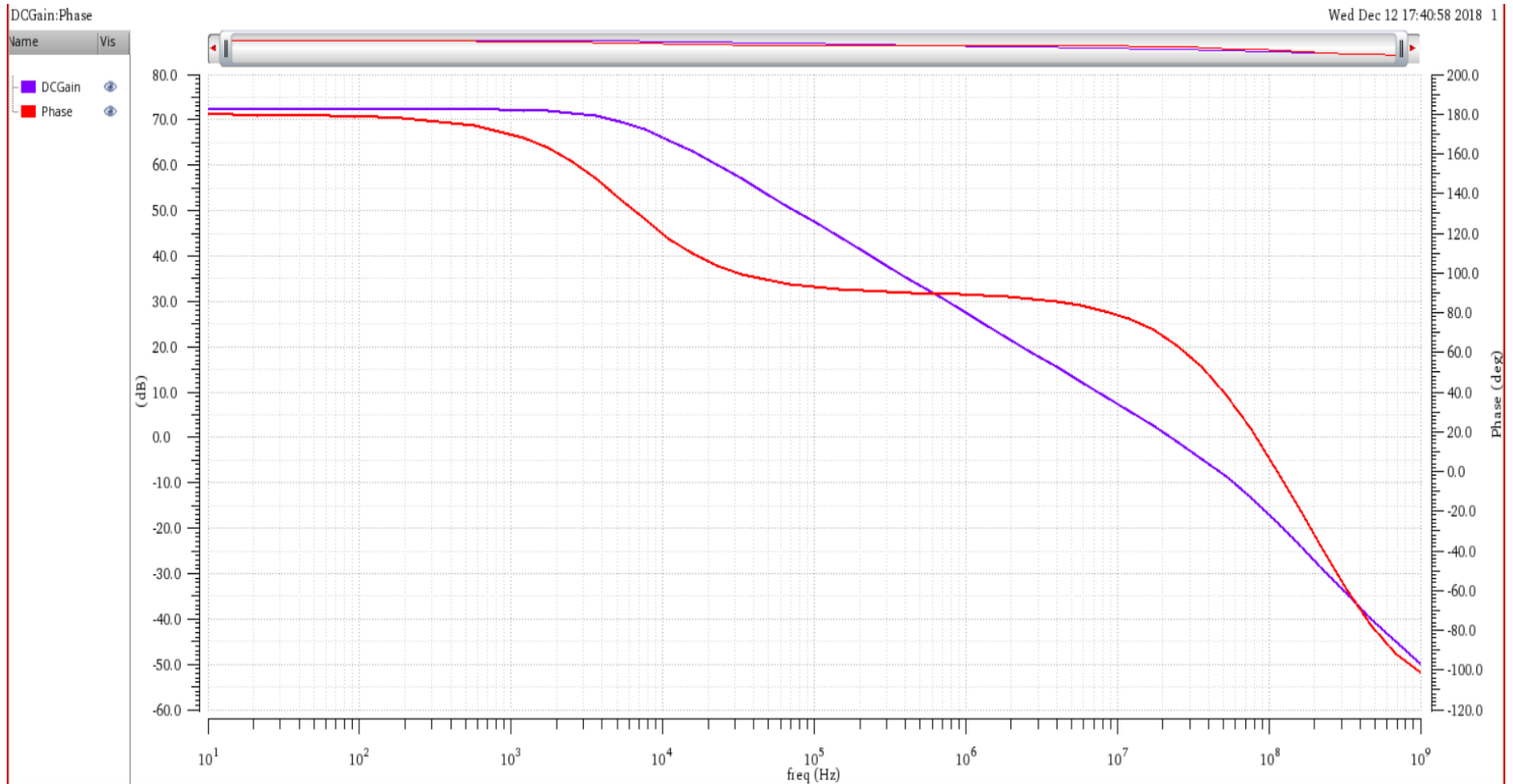


Figure 12. Bode plot of the frequency response of the Miller compensated operational amplifier

Table 3. Miller Compensation Simulation Results

Specification	Miller Compensation
DC Gain	72dB
GBW	23.16 MHz
Phase Margin	65°
Power	290.16 μW
ICMR	0.7 V – 1.7 V
Compensation Capacitor	800 fF

Design Process using Indirect Feedback Compensation Technique:

As explained on the conceptual principles, indirect feedback compensation can be achieved by using an internal low impedance node to indirect fed the compensation current. This can be achieved using a common gate amplifier or a cascode structure. However, the voltage supply level has been scaling down which makes a cascode structure no longer a feasible approach in scaled CMOS processes. As a result, techniques like split-length transistor must be implemented to create a low impedance node to feed the compensation current. Using split-length device, a low impedance node is created since the lower transistor is in triode region which offers a low channel resistance [3]. Indirect feedback frequency compensation can be either obtained by splitting the lengths of the differential pair devices of the load devices. A schematic architecture is shown on figure 13. To better compare the performance of each of the two designs, only the split-length method is incorporated into the two-stage operational amplifier design to create the low impedance node while keeping the same aspect ratios.

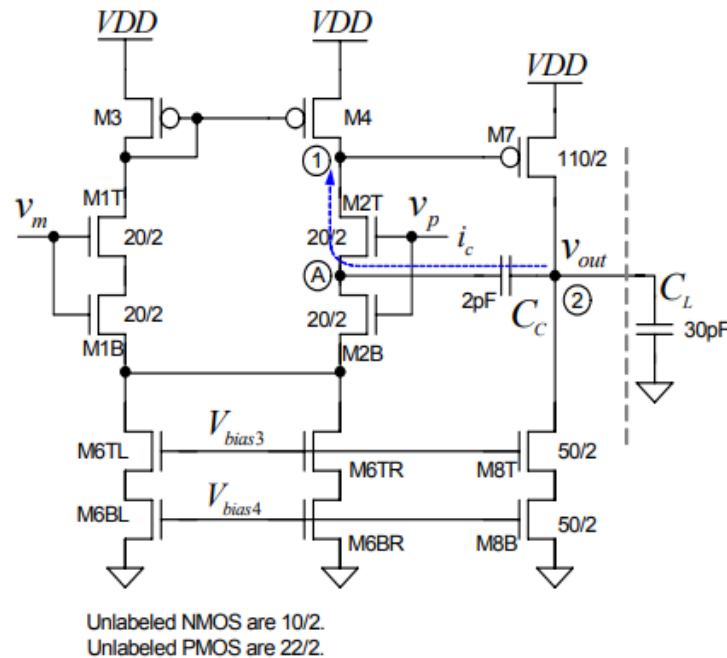


Figure 13. Schematic of indirect feedback compensation technique using split-length

Table IV states the size of each transistor used in the indirect feedback frequency compensation two-stage operational amplifier. Notice that they are the same aspect ratios of the previous designed two-stage op-amp. Since the objective is to compare the performance of each compensation method, keeping the ratio will allow to analyze if indirectly feeding the feedback current exhibits better results than the commonly used direct compensation.

Table 4. Indirect Feedback Compensation Transistor Sizing

Transistor	Aspect Ratio ($\frac{W}{L}$)
M1, M2	6
M3, M4	14
M5	12
M6	173
M7	75

Cadence Design and Simulation of Indirect Feedback Compensation Amplifier

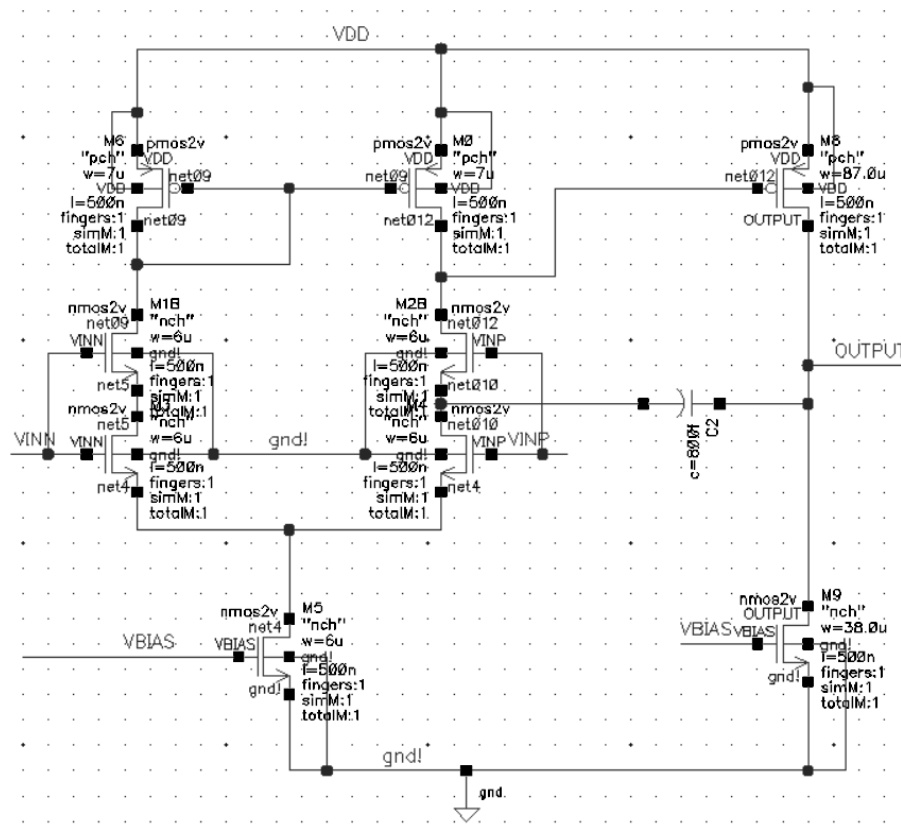


Figure 14. Schematic design of the proposed indirect feedback compensated amplifier

The two-stage operational amplifier with indirect feedback frequency compensation achieved the desired specifications based on the sizing showed on Table IV. The results shown on Table V demonstrates that the performance of the amplifier with an indirect feedback compensation is better than that obtained with direct compensation. As shown below on the frequency response of the operational amplifier, the system approximates the behavior of a single pole system before the unity gain frequency. The only observed drawback is the zero near the unity gain frequency that flatted the gain.

More importantly, the more noticeable parameter is the bandwidth which is more than twice the bandwidth of the direct compensation amplifier. Since the second pole is at a higher frequency, the unity gain frequency is higher. Furthermore, the compensation capacitor is half of the capacitor used with direct Miller compensation, resulting in a small layout area.

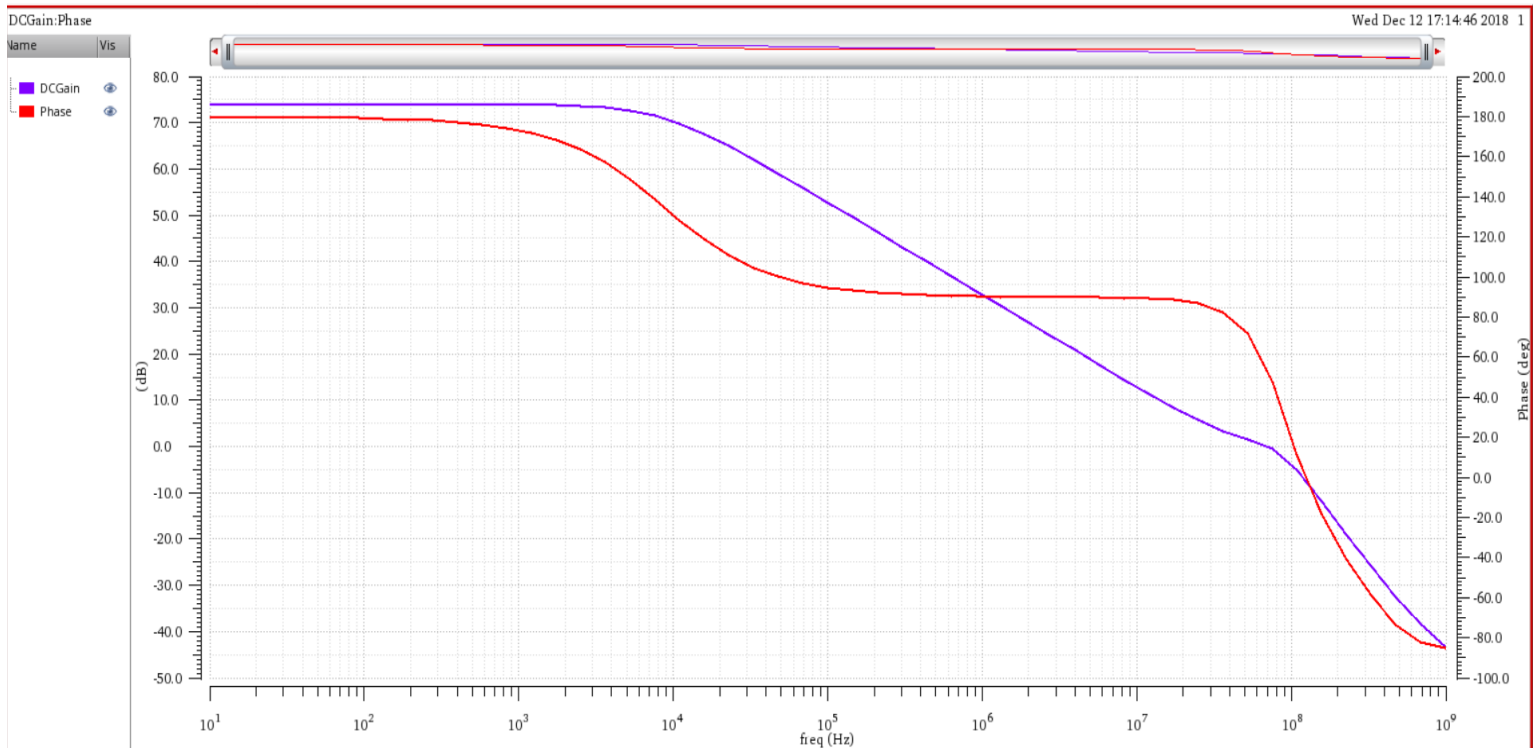


Figure 15. Bode plot of the frequency response of the indirect feedback compensated amplifier

Table 5. Indirect Feedback Compensation Amplifier Results

	Indirect Feedback
DC Gain	<i>74 dB</i>
GBW	<i>60 MHz</i>
Phase Margin	<i>62°</i>
Power	<i>290 μW</i>
ICMR	<i>0.7 V – 1.75 V</i>
Compensation Capacitor	<i>400 fF</i>

Conclusion and Future Work

As integrated circuit system are designed to appear as a single-pole system over a wide frequency range to easy the problem that second order and greater system arises regarding stability, compensation techniques must be improved to meet some design specification constrains like higher unity gain frequency and better phase margin. The comparison demonstrated in this thesis between the Miller compensation technique and the indirect feedback frequency compensation method reflect that this indirect method of compensation reflects potential benefits for providing stability to the operational amplifier. Similarly, this method showed improvement in unity gain frequency and a reduce on the capacitor size. Considering the analysis of this document, indirect feedback compensation shows to be a feasible alternative for compensating amplifiers. Nevertheless, further simulations under specific scenarios should be performed to improve the comparison between these two-compensation techniques. This indirect feedback compensation can be extended to operational amplifier with more than two stages or even different CMOS material processes to explore their advantages against the more commonly used compensation techniques.

Appendix

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